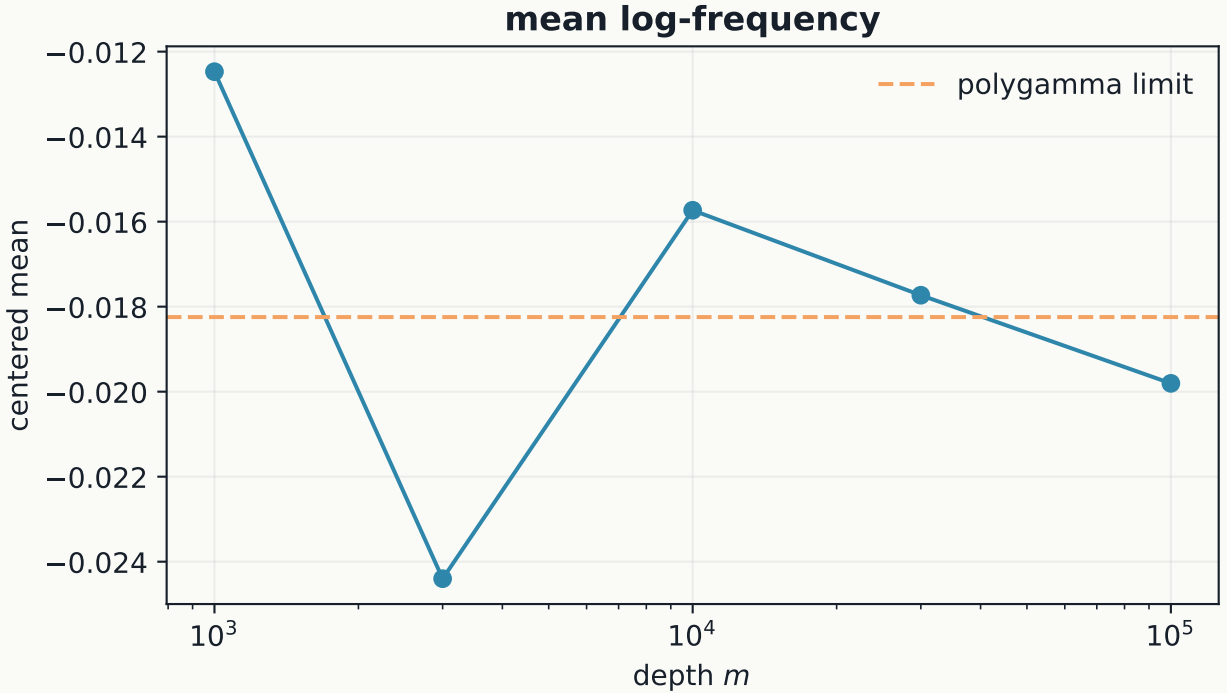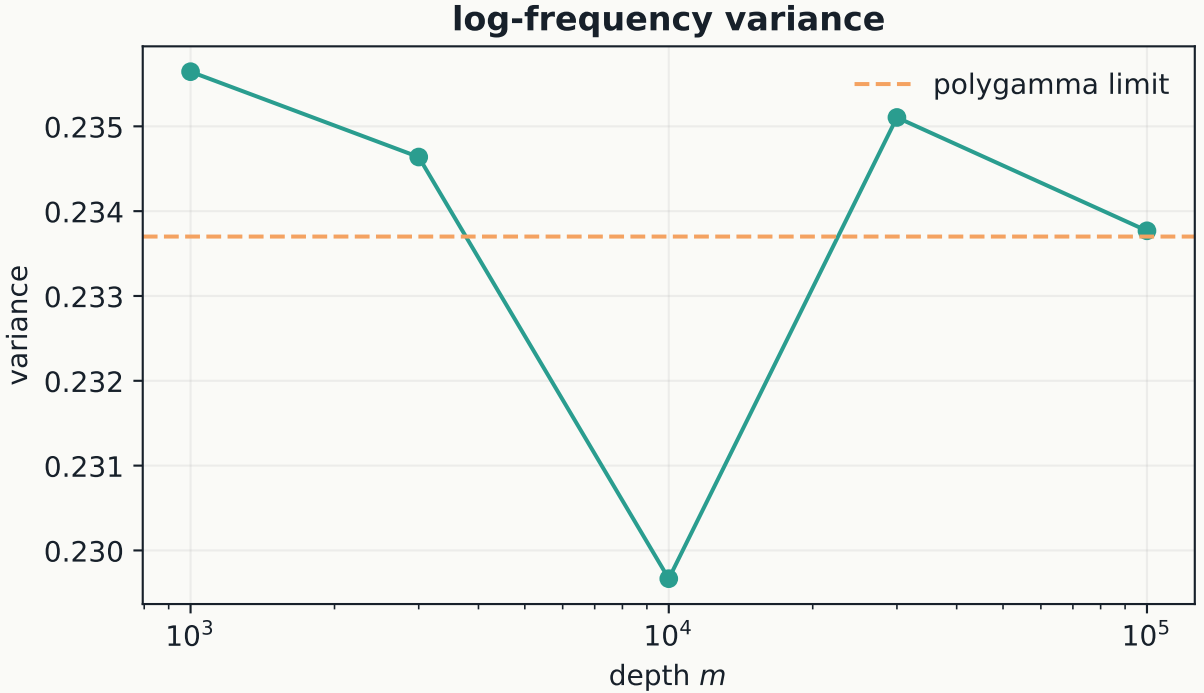Logarithmic spectral cumulants converge to explicit polygamma constants